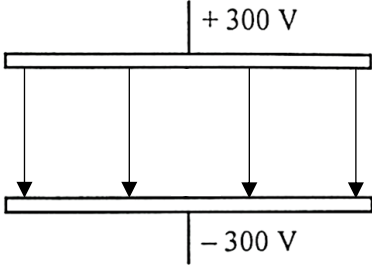


4(a)	Electric field strength is the electric force per unit <u>positive</u> charge at a point, due to an electric field. SI unit is N C^{-1} .	B1 A1
4(b)(i)		A1
4(b)(ii)	Electric field strength = $600 / 0.012$ = 50000 N C^{-1}	A1
4(b)(iii)	Work done = $600 \times 1.6 \times 10^{-19}$ = $9.6 \times 10^{-17} \text{ J}$	A1
4(b)(iv)	gain in kinetic energy = $9.6 \times 10^{-17} \text{ J}$	A1
4(c)	Electric force provides the centripetal force $e^2 / (4 \pi \epsilon_0 r^2) = m_e r \omega^2$ $\omega^2 = (1.6 \times 10^{-19})^2 / [(4 \pi \epsilon_0)(9.11 \times 10^{-31})(5.3 \times 10^{-11})^3]$ $\omega = 4.12 \times 10^{16} \text{ rad s}^{-1}$ Since $T = 2\pi/\omega = 1.53 \times 10^{-16} \text{ s}$ $I = Q/t = 1.6 \times 10^{-19} / (1.53 \times 10^{-16})$ $= 1.05 \times 10^{-3} \text{ A}$	M1 A1 M1 A1